Research Framework: Physical Distinguishability in HCC-S^3

A Preliminary Framework for the Next HCC-S^3 Branch

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1. PURPOSE AND SCOPE

This document is not a research article. It is a research framework. Its purpose is to define the conceptual structure, the central question, the key definitions, and the open problems for a new branch of the Holonomic Compact Cosmology programme.

The branch is called **Physical Distinguishability in HCC-S^3**.

Its focus is narrow: to determine the conditions under which a difference between two quantum states becomes physically real on a compact carrier.

This framework does not:

- Derive information measures.
- Derive the cosmological constant.
- Make claims about consciousness.
- Reference 85/88 levels or special dimensions.
- Claim to solve any existing problem in physics.

It does one thing: it asks what makes a difference physically real, and it proposes a geometric language for answering that question.

2. CENTRAL SHIFT: DISTINGUISHABILITY BEFORE INFORMATION

The classical theory of information, following Shannon, treats information as a measure of uncertainty reduction. The Bekenstein bound treats information as a quantity limited by energy and radius. Both approaches take the existence of distinguishable states as given.

This framework reverses the order.

Thesis. Information is not the starting point. Information is a derived, logarithmic measure of physically realised distinguishability. Before one can count bits, one must determine what makes two states physically distinguishable in the first place.

The conceptual chain is:

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Physical carrier (M, g)
    ↓
Geometric and quantum-geometric constraints
    ↓
Criterion of physical distinguishability
    ↓
Equivalence relation between states
    ↓
RPD(M) — the set of realised physically distinguishable classes
    ↓
N_dist = |RPD(M)|
    ↓
I = log₂ N_dist   (information, derived)
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Each arrow requires justification. The framework does not assume that every arrow can be made rigorous immediately. It identifies which steps are definitions, which are axioms, which are candidate criteria, and which are open problems.

3. THE CENTRAL QUESTION

Under what geometric and quantum-geometric conditions does a difference between two quantum states become physically distinguishable on a compact carrier?

This is not a question about how much information fits in a region. It is a question about what makes information possible at all.

4. DEFINITIONS

4.1 Physical Carrier

Definition 1 (Physical carrier). A physical carrier is a compact Riemannian manifold (M, g) of finite volume, equipped with:

- A Hilbert space H of quantum states defined on M,
- A unitary representation of parallel transport on H,
- A connection defining holonomy and geometric phase.

The total energy E_tot on the carrier is assumed finite and conserved under the dynamics considered.

S^3 is the primary carrier of the present HCC-S^3 branch. However, the formalism is developed for a general compact M so that other manifolds may be tested later.

4.2 Operational Distinguishability

Definition 2 (Physically distinguishable states). Two states $|\psi\rangle$ and $|\varphi\rangle$ in H are physically distinguishable on M if there exists a physically realisable local operation O, implementable with finite energy in finite time, such that

$$|\langle\psi|O|\psi\rangle - \langle\varphi|O|\varphi\rangle| \geq \delta$$

for some fixed threshold $\delta > 0$.

Remarks. This definition is operational. It does not refer to information, entropy, or observers. It refers only to the physical possibility of discrimination by a local operation. The threshold δ is a free parameter of the framework; it may be set by fundamental constants or by experimental context.

4.3 Physical Indistinguishability

Definition 3 (Physically indistinguishable states). Two states $|\psi\rangle$ and $|\varphi\rangle$ in H are physically indistinguishable on M if no physically realisable local operation O can discriminate them with fidelity $\geq \delta$.

Remarks. This is the negation of Definition 2. It defines an equivalence relation $\sim$ on H:

$$|\psi\rangle \sim |\varphi\rangle \;\Leftrightarrow\; |\psi\rangle \text{ and } |\varphi\rangle \text{ are physically indistinguishable on M.}$$

4.4 Realised Physical Distinguishability Set

Definition 4 (RPD(M)). The set RPD(M) is the quotient space

$$\mathrm{RPD}(M) = H / \sim,$$

where $\sim$ is the equivalence relation of physical indistinguishability.

Each element of RPD(M) is an equivalence class of states that cannot be physically distinguished from one another on M. Distinct elements of RPD(M) correspond to physically distinguishable classes.

4.5 Derived Quantities

Definition 5 (N_dist). N_dist = |RPD(M)|, the cardinality of the set of realised physically distinguishable classes.

Definition 6 (Information, derived). The physical information capacity of the carrier is

$$I(M) = \log_2 N_dist,$$

expressed in bits. This is a logarithmic measure of the number of physically distinguishable classes. It is derived from RPD(M), not fundamental.

5. GEOMETRIC FUNCTIONAL $\mathcal{G}(M)$

To connect distinguishability to geometry, we introduce a general geometric functional $\mathcal{G}(M)$.

Definition 7 (Geometric functional). $\mathcal{G}(M)$ is the collection of all geometrically and topologically invariant data on M that may constrain or enable physical distinguishability. Its components include, when relevant:

- Hol(M): the holonomy group of the connection,
- Riem(M): curvature invariants (Ricci scalar, sectional curvature, etc.),
- Spec(Δ): the spectrum of the scalar Laplacian,

- Spec(D): the spectrum of the Dirac operator, if M admits a spin structure,
- Spin(M): the spin structure itself,
- Δ_E: the spectral gap between the ground state and the first excited state,
- Berry(M): the structure of Berry phases for adiabatic transport along closed paths,
- Conn(M): the connection data defining parallel transport.

$\mathcal{G}(M)$ is not claimed to be a complete invariant. It is a working collection of the geometric data that are candidates for constraining distinguishability.

Candidate Criterion (Geometric constraint on distinguishability). Two states are physically distinguishable on M only if they differ in their response to at least one element of $\mathcal{G}(M)$. That is, indistinguishability implies identical geometric phases, identical spectral projections, and identical holonomy transformations for all physically accessible closed paths.

This criterion is a conjecture. It is not proved. It may be too strong. The framework treats it as a research hypothesis to be tested.

6. AXIOMS

The framework rests on a small number of axioms. These are not derivable from more fundamental principles within the framework; they define its domain.

Axiom 1 (Carrier finiteness). The physical carrier (M, g) is compact, of finite volume, and supports a finite total energy E_tot.

Axiom 2 (Operational discriminability). The only criterion for physical distinguishability is the existence of a local operation O satisfying Definition 2. There is no distinction without operational difference.

Axiom 3 (Geometric grounding). The physical distinguishability or indistinguishability of states on M is constrained by the geometric structure of M as encoded in $\mathcal{G}(M)$. Two states that are identical with respect to all elements of $\mathcal{G}(M)$ are candidates for physical indistinguishability.

Axiom 4 (Finite distinguishability). The set RPD(M) has finite cardinality. N_dist is finite.

Remarks. Axiom 4 is consistent with the Bekenstein bound but does not assume it. Finiteness follows from the compactness of M, the finiteness of E_tot, and the operational nature of distinguishability.

7. RESEARCH QUESTIONS

The framework generates the following research questions. They are ordered from foundational to derived.

RQ1 (Equivalence relation consistency). Is physical indistinguishability as defined in Definition 3 an equivalence relation on H? Or does the operational definition need refinement to ensure transitivity?

RQ2 (Geometric characterisation of ~). Can the equivalence relation $\sim$ be characterised purely in terms of $\mathcal{G}(M)$? That is, are two states physically indistinguishable if and only if they have identical invariants under all elements of $\mathcal{G}(M)$?

RQ3 (Cardinality of RPD(M)). What is $|\mathrm{RPD}(M)|$ for $M = S^3$ with the round metric, the spin connection, and the standard Dirac operator? Can it be computed from $\mathcal{G}(S^3)$?

RQ4 (Holonomy as obstruction). Is holonomy a necessary component of the geometric constraint on distinguishability? Or can two states be physically distinguishable even if they have identical holonomy phases for all closed paths?

RQ5 (Spectral gap and distinguishability). Does the spectral gap Δ_E play a privileged role? Is there a minimum energy difference required for two states to be physically distinguishable?

RQ6 (Relation to Bekenstein). Does the geometric bound on |RPD(M)| coincide with, constrain, or exceed the Bekenstein bound? Are they independent, complementary, or equivalent?

8. OPEN PROBLEMS

The following open problems are more specific and more technical than the research questions. They define concrete mathematical tasks.

OP1 (Formalise the operational definition). Refine Definition 2 into a mathematically rigorous criterion. Specify the class of admissible operations O. Define the threshold δ in terms of fundamental constants or operational context.

OP2 (Prove or disprove the equivalence relation property). Prove or find a counterexample to the transitivity of physical indistinguishability as defined in Definition 3. If transitivity fails, repair the definition.

OP3 (Compute $\mathcal{G}(S^3)$). Compile the complete geometric functional for S^3 with the round metric, the spin connection, and the Dirac operator. Identify which components of $\mathcal{G}(S^3)$ are non-trivial and may constrain distinguishability.

OP4 (Berry phase structure on S^3). Compute the Berry phases for adiabatic transport of spinor states along closed geodesics on S^3. Determine whether states with identical Berry phases can be physically distinguished by other operations.

OP5 (Lower bound on N_dist). Derive a lower bound on N_dist for S^3 from $\mathcal{G}(S^3)$. This would show that the geometric structure not only limits but also guarantees a minimum number of distinguishable states.

OP6 (Relation to fractal re-quantisation). Connect the distinguishability classes RPD(S^3) to the fractal re-quantisation mechanism of Preece and Batenin (2026c). Does each level of the φ-hierarchy correspond to a new distinguishability class?

9. METHODOLOGICAL DISCIPLINE

This framework imposes strict methodological rules on any research conducted under its heading.

1. **Distinguishability is primary.** No argument may treat information as fundamental. Information is always $\log_2$ of a count of distinguishable classes.
2. **Operational definitions only.** No state is declared distinguishable on the basis of mathematical orthogonality alone. An operation O must be specified.
3. **Geometry constrains, not dictates.** $\mathcal{G}(M)$ provides candidate constraints. Whether a specific element of $\mathcal{G}(M)$ actually constrains distinguishability is a research question, not an assumption.
4. **No cosmological shortcuts.** This framework does not derive Λ, does not predict Hubble constants, and does not explain dark energy. Those are separate branches of HCC.
5. **S^3 is a test case.** The primary carrier is S^3, but the framework must be formulated so that $M = T^3$, $M = S^1 \times S^2$, and other compact manifolds can be tested as alternatives.
6. **Separation of labels.** Every statement must be labelled as one of: Definition, Axiom, Candidate Criterion, Research Question, Open Problem, Conjecture. Claims stronger than Conjecture require proof.

10. RELATION TO PREVIOUS HCC WORK

This framework depends on three prior HCC results:

- **Causal coherence on S^3** (Preece and Batenin, 2026a): establishes that S^3 is a well-behaved compact causal carrier with finite t_coh.
- **Physical carrier concept** (Preece and Batenin, 2026d): establishes the distinction between geometric and physical carriers and identifies holonomy as a candidate constraint.
- **Energy density and distinguishability** (Preece and Batenin, 2026e): separates energy density, state density, distinguishability, and information density, and introduces the redistribution axiom.

The present framework takes the physical carrier concept from (2026d) and the conceptual separations from (2026e) and pushes deeper: it asks not what a physical carrier is, but what makes distinguishability physically real on that carrier.

11. SUMMARY OF THE FRAMEWORK

Central thesis. Information is the logarithmic measure of realised physical distinguishability. Distinguishability is therefore prior to information.

Central question. Under what geometric and quantum-geometric conditions does a difference between two states become physically distinguishable on a compact carrier?

Method. Define operational distinguishability. Define the equivalence relation of indistinguishability. Form the quotient set RPD(M). Count its elements. Derive information as the logarithm of that count.

Geometric input. $\mathcal{G}(M)$, a collection of geometric invariants including holonomy, curvature, spectra, spin structure, and Berry phase data.

Output. A research programme with 6 research questions and 6 open problems, constrained by 6 methodological rules.

Status. This is a framework, not a theory. It makes no empirical claims. It provides a structure within which empirical claims may eventually be formulated and tested.

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